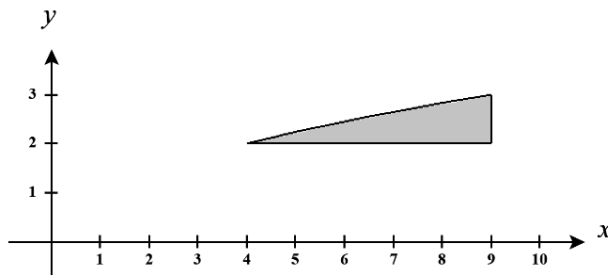


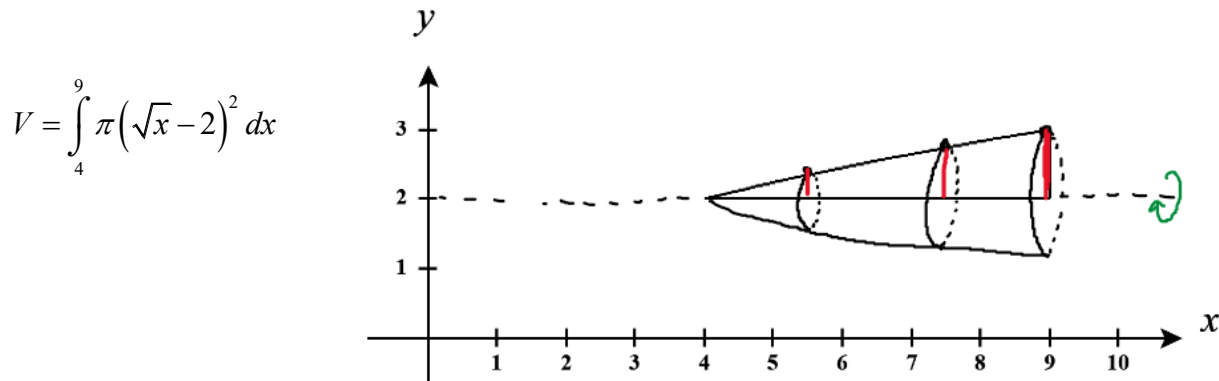
Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

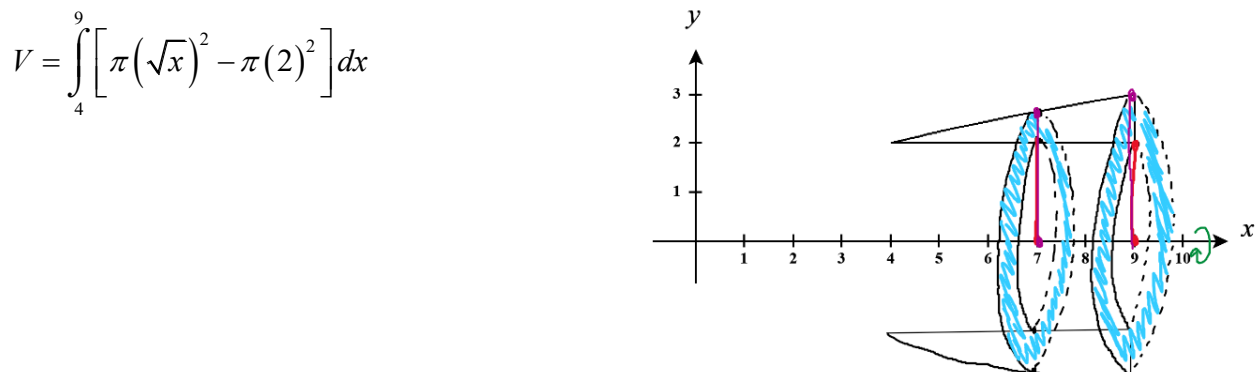
1. Consider the region R determined by the curves $y = \sqrt{x}$ & $y = 2$ on the interval $[4, 9]$.



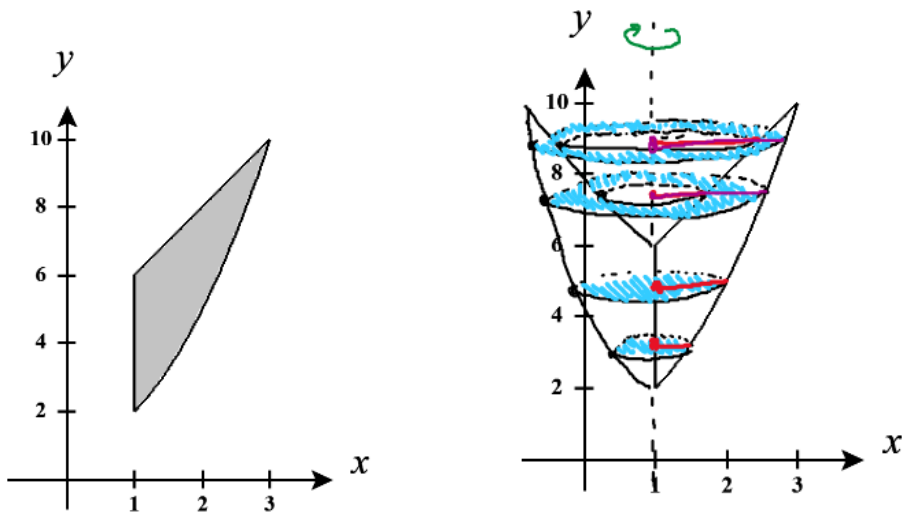
- a) Set up, but do not evaluate, an integral which represents the volume of the solid when R is revolved around the line $y = 2$. Include a sketch of your solid and cross sections.



- b) Set up, but do not evaluate, an integral which represents the volume of the solid when R is revolved around the x -axis. Include a sketch of your solid and cross sections.



2. Consider the region R determined by the curves $y = x^2 + 1$, $y = 2x + 4$, & $x = 1$ and that lies to the right side of $x = 1$. Set up, but do not evaluate, an integral which represents the volume of the solid when R is revolved around the line $x = 1$. Include a sketch of your solid and cross sections.

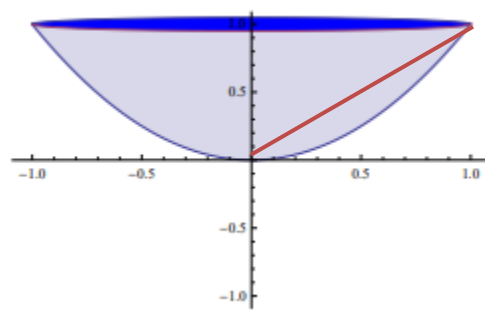
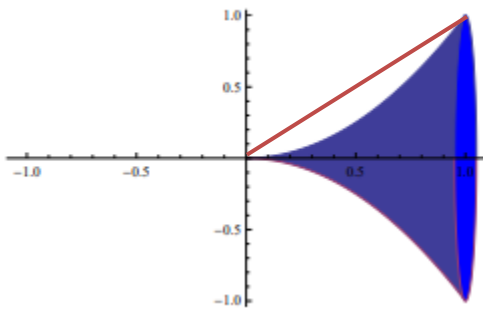


$$V = \int_2^6 \left[\pi (\sqrt{y-1} - 1)^2 \right] dy + \int_6^{10} \left[\pi (\sqrt{y-1} - 1)^2 - \pi \left(\frac{1}{2}(y-4) \right)^2 \right] dy$$

3. How can you tell whether you will need to use washers or disks to assist in finding the volume of a solid of rotation?

The key is to look at whether or not there is a gap between the region you are rotating and the axis of rotation.

4. The region under the graph of $y = x^2$, $0 \leq x \leq 1$, is rotated about the x and y axes. The graphs below illustrate the solids of revolution which are formed.



- a) Set up and evaluate the integral giving the volume of the solid obtained by rotation about the x -axis.

$$V = \int_0^1 \pi (x^2)^2 dx = \int_0^1 \pi x^4 dx = \frac{\pi}{5} x^5 \Big|_0^1 = \frac{\pi}{5} (1) - \frac{\pi}{5} (0) = \frac{\pi}{5}$$

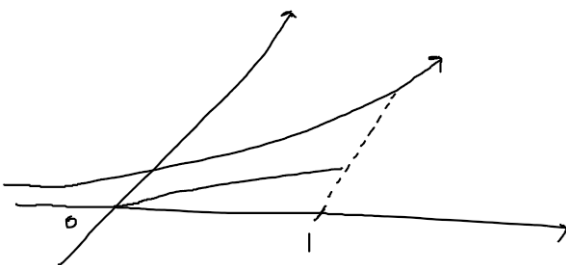
- b) Without calculating the volume of the other solid determine which solid has a larger volume. (Note that saying “the second figure looks bigger” is not a convincing argument.)

We could note that if we drew a line segment connecting the origin to $(1,1)$ we can see that by rotating this line around each axis would produce a volume that we can then compare to the given solids.

Note that in both pictures we create a cone when rotating the red line around the axis. In both pictures the volume of this cone is given by $V = \frac{1}{3} \pi (1)^2 (1) = \frac{\pi}{3}$ and therefore we can see that first solid must

have a smaller volume and the second solid must have a larger volume than $\frac{\pi}{3}$. Thus, the second solid must have a larger volume.

5. Consider a solid built onto of the region determined by $y = e^x$, $y = x$, $x = 0$, and $x = 1$.



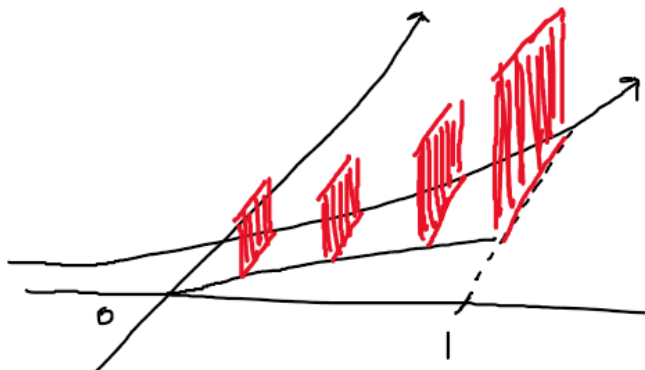
- a) Setup, but do not evaluate, an integral which represents the volume of the solid if the cross sections for this solid which are parallel to the y -axis are squares.

The area of each cross section is given by

$$A = (e^x - x)^2.$$

Therefore, the volume of the solid is

$$V = \int_0^1 (e^x - x) dx$$



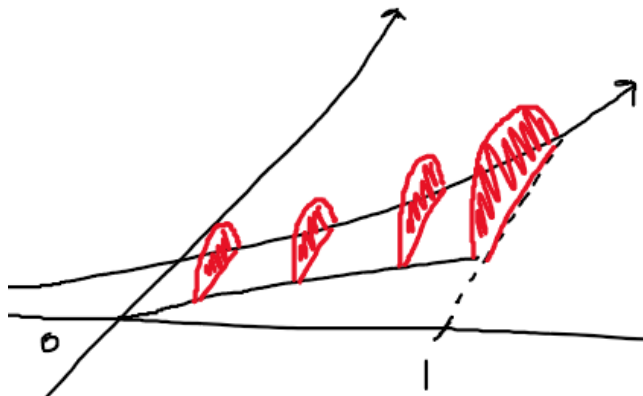
- b) Setup, but do not evaluate, an integral which represents the volume of the solid if the cross sections for this solid which are perpendicular to the x -axis are semi-circles.

The area of each cross section is given by

$$A = \frac{\pi}{2} \left(\frac{e^x - x}{2} \right)^2.$$

Therefore, the volume of the solid is

$$V = \int_0^1 \frac{\pi}{2} \left(\frac{e^x - x}{2} \right)^2 dx$$



- c) Setup, but do not evaluate, an integral which represents the volume of the solid if the cross sections for this solid which are perpendicular to the y -axis are rectangles which are twice as tall as they are wide.

The area of each cross section is not always given by the same formula.

On $0 \leq y \leq 1$ the area of the cross section is given

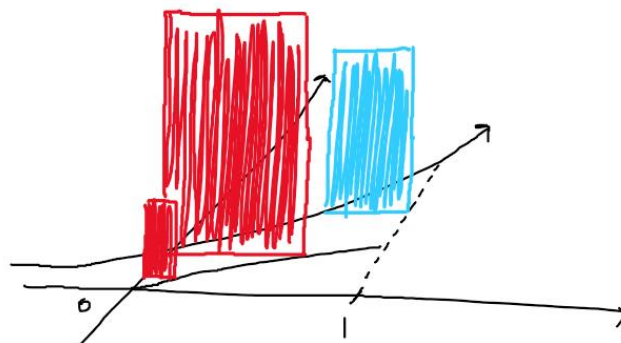
$$A = 2(y - 0)^2.$$

On $1 \leq y \leq e$ the area of the cross section is given

$$A = 2(1 - \ln(y))^2.$$

Therefore, the volume of the solid is

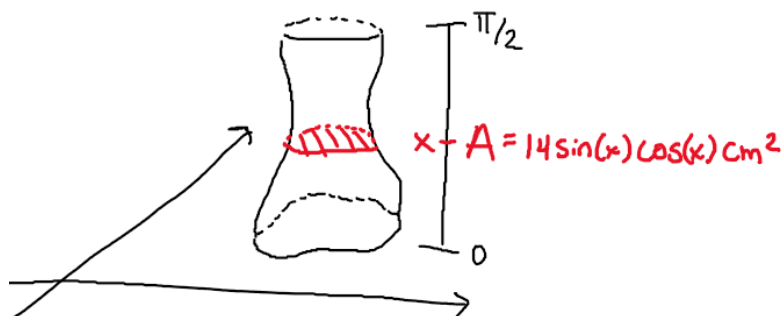
$$V = \int_0^1 2y^2 dy + \int_1^e 2(1 - \ln(y))^2 dy$$



6. Consider a three-dimensional shape standing on the floor with height $\frac{\pi}{2}$ cm such that the area of a cross section parallel to the floor is $14\sin(x)\cos(x) \text{ cm}^2$ for any height $0 \leq x \leq \frac{\pi}{2}$ cm. What is the volume of the three-dimensional shape?

Note that we have already been given a formula for the volume of each cross section and therefore we can easily compute the volume by the following.

$$\begin{aligned} V &= \int_0^{\pi/2} 14\sin(x)\cos(x) dx = \int_0^1 14u du \\ &= 7u^2 \Big|_0^1 = 7 - 0 \\ &= 7 \end{aligned}$$

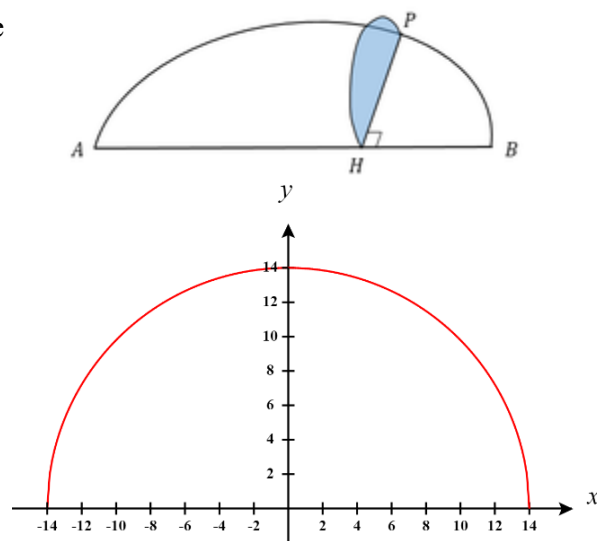


7. Consider the three-dimension object whose base is a semicircle with diameter $AB = 28$. If each cross section of the three-dimension which are perpendicular to $\overline{AB}$ are also semicircles, then what is the volume of the resulting object?

We first move this scenario onto the xy -plane. We see that the circle here is given by the equation $x^2 + y^2 = 14^2$ and therefore the curve here is given by $y = \sqrt{14^2 - x^2}$.

So, the volume of the solid is given by

$$\begin{aligned} V &= \int_{-14}^{14} \frac{1}{2} \pi \left(\frac{\sqrt{196 - x^2}}{2} \right)^2 dx = \int_{-14}^{14} \frac{1}{2} \pi \left(\frac{196 - x^2}{4} \right) dx \\ &= \int_{-14}^{14} \frac{1}{8} \pi (196 - x^2) dx = \frac{\pi}{8} \left(196x - \frac{1}{3}x^3 \right) \Big|_{-14}^{14} \\ &= \frac{\pi}{8} \left(196(14) - \frac{1}{3}(14)^3 \right) - \frac{\pi}{8} \left(196(-14) - \frac{1}{3}(-14)^3 \right) \\ &= \frac{\pi}{8} \left(2744 - \frac{2744}{3} \right) - \frac{\pi}{8} \left(-2744 + \frac{2744}{3} \right) \\ &= 686\pi \left(\frac{2}{3} \right) \\ &= \frac{1372\pi}{2} \end{aligned}$$



8. The yellow-colored region in the diagram below is bounded by $y = 18\sin(x) - a$, $x = 0$, $x = \pi$, & $y = 0$, where a is a real number such that $0 \leq a \leq 18$. What is the value of a that minimizes the volume of the solid obtained by rotating the region around the x -axis?

We see that the volume of the given solid would be

$$V = \int_0^{\pi} \pi [18\sin(x) - a]^2 dx = \pi \int_0^{\pi} [324\sin^2(x) - 36a\sin(x) + a^2] dx$$

$$= \pi \int_0^{\pi} \left[\frac{324}{2}(1 - \cos(2x)) - 36a\sin(x) + a^2 \right] dx$$

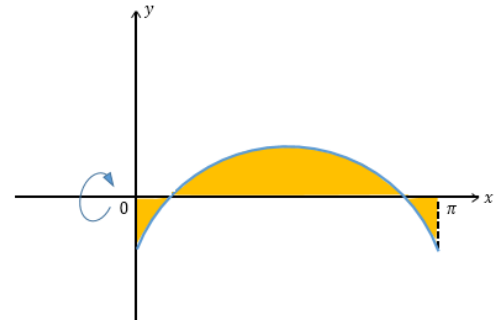
$$= \pi \left[162 \left(x - \frac{1}{2} \sin(2x) \right) + 36a \cos(x) + a^2 x \right]_0^{\pi}$$

$$= \pi \left[162 \left(\pi - \frac{1}{2} \sin(2\pi) \right) + 36a \cos(\pi) + a^2 \pi \right] - \pi \left[162 \left(0 - \frac{1}{2} \sin(0) \right) + 36a \cos(0) + a^2 (0) \right]$$

$$= \pi [162(\pi - 0) - 36a + a^2 \pi] - \pi [36a]$$

$$= 162\pi^2 - 36a\pi + a^2\pi^2 - 36a\pi$$

$$= 162\pi^2 - 72a\pi + a^2\pi^2$$

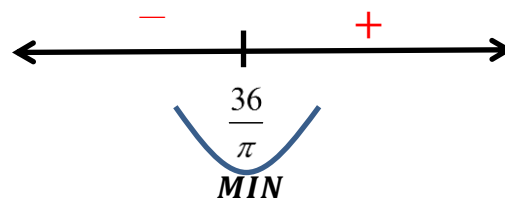


Therefore, the volume of the solid is given by $V = 162\pi^2 - 72a\pi + a^2\pi^2$. We want to find the value of a that minimizes the volume of the solid obtained, so we must use calculus I techniques to determine where the minimum occurs.

$$V' = 0 - 72\pi - 2a\pi^2$$

$$V' = 0 \text{ when } -72\pi - 2a\pi^2 = 0 \rightarrow a = \frac{72\pi}{2\pi^2} = \frac{36}{\pi}$$

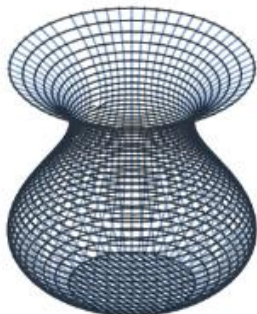
V' is never undefined.



We can then see that when $a = \frac{36}{\pi}$ the volume of the solid is minimized.

9. A vase maker decides to construct a vase whose contour is a cubic polynomial according to the following specifications:

- The vase is 4 feet tall.
- The vase is 4 feet in diameter at its widest (which occurs 1 foot from the base).
- The vase is 2 feet in diameter at its narrowest (which occurs 1 foot from the top).
- The vase has a flat circular bottom.



Rounded to 3 decimal places, what is the volume of the vase in cubic feet?

We need to determine the polynomial function that determines this vase. If we lay the solid on its side (like in the picture below) and imagine that the left side starts at $x = 0$, then right sides ends at $x = 4$.

We see that the polynomial that we need has the form of

$$f(x) = ax^3 + bx^2 + cx + d \text{ and that}$$

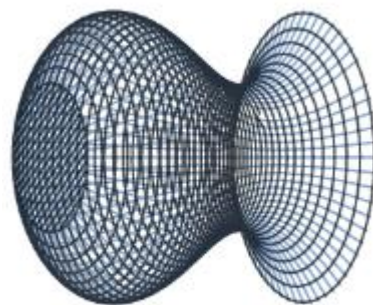
$$f(1) = 2, \text{ which implies } 2 = a + b + c + d$$

$$f(3) = 1, \text{ which implies } 1 = 27a + 9b + 3c + d$$

Note that $f'(x) = 3ax^2 + 2bx + c$ and so

$$f'(1) = 0, \text{ which implies } 0 = 3a + 2b + c \text{ (since there is a maximum height here).}$$

$$f'(3) = 0, \text{ which implies } 0 = 27a + 6b + c \text{ (since there is a maximum height here).}$$



So, we have a system of equations

$$\begin{cases} 2 = a + b + c + d \\ 1 = 27a + 9b + 3c + d \\ 0 = 3a + 2b + c \\ 0 = 27a + 6b + c \end{cases} \rightarrow \begin{cases} a = \frac{1}{4} \\ b = -\frac{3}{2} \\ c = \frac{9}{4} \\ d = 1 \end{cases}$$

Therefore, we have that $f(x) = \frac{1}{4}x^3 - \frac{3}{2}x^2 + \frac{9}{4}x + 1$ and the volume of the vase is then given by

$$V = \int_0^4 \pi [f(x)]^2 dx = \int_0^4 \pi \left[\frac{1}{4}x^3 - \frac{3}{2}x^2 + \frac{9}{4}x + 1 \right]^2 dx . \text{ Using our calculator, we find that}$$

$$\int_0^4 \pi \left[\frac{1}{4}x^3 - \frac{3}{2}x^2 + \frac{9}{4}x + 1 \right]^2 dx \approx 29.8 ft^3$$